Esercitazione 6

First-Principles NMR Chemical Shift Estimation and Thermochemical Analysis

Geometry Optimization and the Potential Energy Surface

Before computing any molecular property, such as an NMR spectrum, the molecule must be relaxed into its ground-state equilibrium geometry. Under the **Born–Oppenheimer approximation**, the rapid motion of light electrons is decoupled from the sluggish motion of heavy nuclei. This allows us to define a Potential Energy Surface (PES), $E(\mathbf{R})$, where the electronic energy is a function of the nuclear coordinates $\mathbf{R}$.

Geometry optimization algorithms systematically adjust internal coordinates, such as bond lengths, angles, and dihedral angles, to locate a local minimum on the PES. A mathematical stationary point is reached when the energy gradient, corresponding to the forces acting on the atoms, is zero:

$$\nabla E = 0$$

The Physics of Nuclear Magnetic Resonance

NMR spectroscopy exploits the quantum spin properties of atomic nuclei, such as ^{1}H and ^{13}C. When placed in a strong, external, static magnetic field, $\mathbf{B}_0$, the degenerate nuclear spin states undergo Zeeman splitting.

However, a nucleus embedded within a molecule does not "feel" the exact external field. The surrounding electron cloud circulates in response to $\mathbf{B}_0$, generating a small, local induced magnetic field, $\mathbf{B}_{\text{ind}}$, that classically opposes the applied field according to Lenz's law. The effective magnetic field felt by the nucleus is:

$$\mathbf{B}_{\text{eff}} = \mathbf{B}_0 - \mathbf{B}_{\text{ind}} = \mathbf{B}_0 \left(\mathbf{1} - \boldsymbol{\sigma}\right)$$

where $\boldsymbol{\sigma}$ is the **magnetic shielding tensor**, a 3×3 matrix. In isotropic fluid phases, such as liquids or gases, rapid molecular tumbling mathematically averages this tensor to its trace, yielding a single scalar value known as the **isotropic shielding**, σ_{iso}.

The GIAO Method

Calculating $\boldsymbol{\sigma}$ quantum-mechanically requires defining a vector potential for the external magnetic field. In practical computations using finite basis sets, artificially shifting the origin of this coordinate system unphysically alters the computed magnetic properties. This flaw is known as the **gauge-origin problem**.

To rigorously solve this, we employ the **GIAO**, or **Gauge-Including Atomic Orbital**, method. GIAO attaches a complex phase factor to each atomic orbital basis function that depends explicitly on the external magnetic field and the spatial position of the atom. This guarantees that the computed NMR shielding tensors are strictly independent of the coordinate system's origin.

The Chemical Shift

Quantum chemistry calculates the *absolute* magnetic shielding, σ_{iso}. Experimental NMR instruments, however, measure the **chemical shift**, δ, which is defined relative to a standard reference compound. For ^{1}H, this is universally **tetramethylsilane**, or TMS, with formula $\text{Si}(\text{CH}_3)_4$.

The chemical shift, expressed in parts per million, ppm, is calculated as:

$$\delta_{\text{sample}} = \sigma_{\text{iso}}^{\text{TMS}} - \sigma_{\text{iso}}^{\text{sample}}$$

Crucial Rule: To ensure a cancellation of systematic computational errors, both $\sigma_{\text{iso}}^{\text{TMS}}$ and $\sigma_{\text{iso}}^{\text{sample}}$ **must** be computed using the exact same theoretical method and basis set.

Practical Workflow

Task Assignment

The objective is to compute the ^{1}H NMR shift for the following molecules:

Acetic acid, ethanol, acetaldehyde, dimethyl ether, acetone.

Rule: Each student must perform calculations for a single molecule. Coordinate within your group to ensure you do not calculate the same target.

Execution Steps

1. **Geometry construction:** Launch **Molden**. Use the **Zmat editor tool** to mathematically construct the 3D geometry of the standard reference, TMS. A Z-matrix defines the geometry using internal coordinates, which optimizes much faster than Cartesian coordinates. Save the Z-matrix. Repeat this for your assigned molecule.
2. **Optimization:** Run a geometry optimization and frequency calculation for TMS.
3. **NMR calculation:** Using the converged, optimized TMS geometry, run a single-point NMR calculation using the GIAO method.
4. **Target calculation:** Repeat Steps 2 and 3 for your assigned target molecule.
5. **Data extraction:** Extract the isotropic ^{1}H magnetic shielding, σ_{iso}, for both TMS and your molecule. Compute the shift, δ.
6. **Validation:** Compare your theoretical results with experimental data reported in the ChemicalBook.

Gaussian Input Examples

We will use the **B3LYP** density functional paired with the **6-311+G(d,p)** basis set (as an alternative, you can use the Def2TZVP basis set). The inclusion of diffuse, `+`, and polarization, `d,p`, functions is scientifically critical to accurately describe the electron density tails far from the nucleus, which heavily influence magnetic shielding.

Example 1: Geometry Optimization and Frequencies

It is mandatory scientific practice to perform a frequency calculation, `freq`, alongside optimization, `opt`, to mathematically guarantee the structure is a true energy minimum.

File: `tms_opt.com`

```
%nprocshared=24
%mem=300Gb
# b3lyp/6-311+g(d,p) opt freq

TMS Geometry Optimization and Frequencies

0 1
 [PASTE YOUR MOLDEN Z-MATRIX OR CARTESIAN COORDINATES HERE]
```

Example 2: NMR Calculation

Once the optimization terminates successfully, indicated by "Normal termination of Gaussian", you must pass the minimized coordinates to the NMR job. Copy the optimized geometry in the new input for the NMR calculation and proceed with a freq and an NMR calculation.

File: `tms_nmr.com`

```
%nprocshared=24
%mem=300GB
# b3lyp/6-311+g(d,p) Freq nmr=giao int=ultrafine

TMS GIAO NMR Calculation

0 1
```

Analyzing the NMR Output

Open the NMR output file, `.log` or `.out`. Search for the string:

```
SCF GIAO Magnetic shielding tensor (ppm)
```

Locate the `Isotropic =` value corresponding to the hydrogen atoms.

- **For TMS:** Because of minor numerical noise, the 12 equivalent protons might vary by fractions of a ppm. Calculate the **arithmetic average** of all 12 protons to find your exact $\sigma_{\text{iso}}^{\text{TMS}}$.
- **For your assigned molecule:** Remember to average the shielding values only for chemically equivalent protons; for example, average the three protons of a methyl group together to account for rapid room-temperature bond rotation.

Advanced NMR Spectrum Interpretation: Spin–Spin Splitting and Spectral Simulation

To fully understand and visualize an NMR spectrum, you must transition from calculating a single scalar value, namely the isotropic chemical shift, to understanding how neighboring nuclei magnetically interact to produce **multiplet splittings**.

Theoretical Background: Spin–Spin Splitting and J-Coupling

While the **chemical shift**, δ, is dictated by the local electron cloud shielding the nucleus from the external magnetic field, $\mathbf{B}_0$, **spin–spin splitting** arises from magnetic

interactions between neighboring NMR-active nuclei, such as two adjacent ^{1}H atoms.

In isotropic liquid solutions, direct through-space magnetic interactions are mathematically averaged to zero by rapid molecular tumbling. Therefore, the splitting observed in conventional liquid-state NMR is transmitted **indirectly through the bonding electrons**.

The dominant quantum mechanical mechanism responsible for scalar spin–spin coupling is the **Fermi contact term**. This contribution requires the electron to have a non-zero probability of being physically located exactly at the nucleus, a property associated primarily with s-orbital character. The nuclear spin polarizes the s-electron, which transmits this polarization through the molecular orbital framework to a neighboring nucleus.

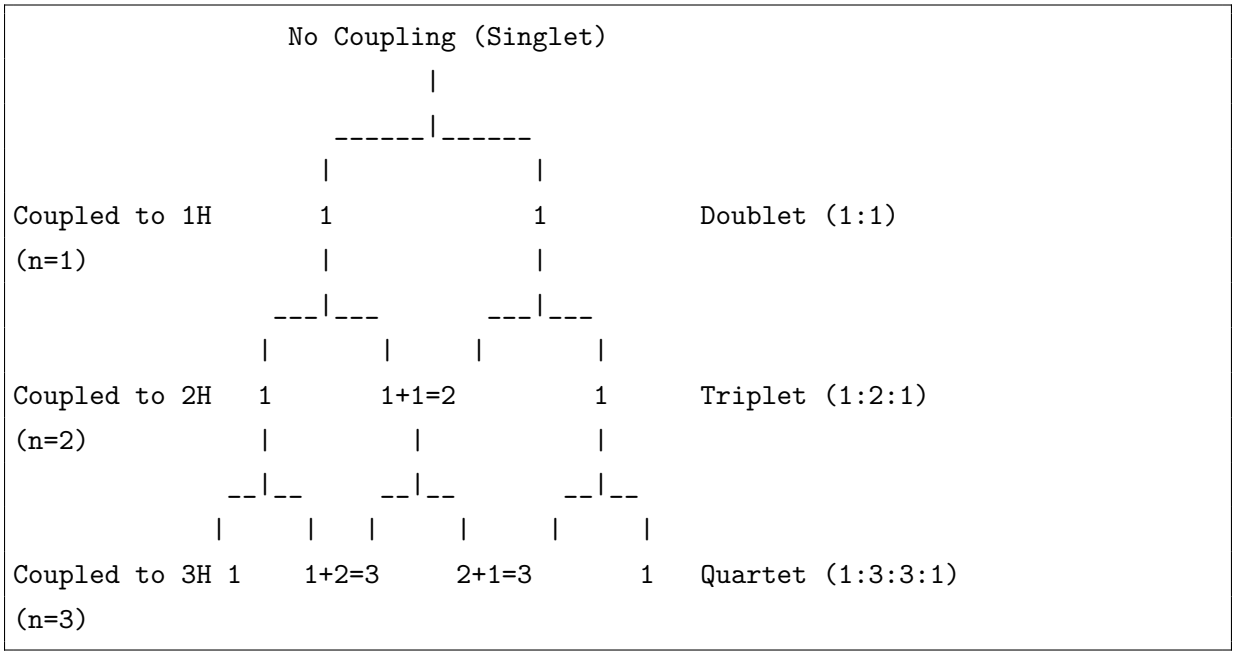
The $n+1$ Splitting Rule and Pascal's Triangle

For simple, first-order aliphatic systems, such as ethanol and dimethyl ether, the splitting follows the $n+1$ rule:

A proton coupled to n equivalent neighboring protons is split into $n+1$ sub-peaks.

Unlike the chemical shift, which scales with the spectrometer magnetic field strength, the ***J*-coupling constant** is measured in Hertz, Hz, and is an intrinsic molecular property. It is strictly independent of the external magnetic field strength.

The following schematic illustrates how the relative intensities of the sub-peaks mirror **Pascal's Triangle**:



Computing J-Couplings in Gaussian

To theoretically obtain the splitting constants, J-values in Hz, the standard GIAO calculation is insufficient. Gaussian must be explicitly instructed to compute the indirect spin–spin coupling matrix.

Warning for Process Engineering Students: Computing spin–spin coupling requires evaluating second derivatives of the electronic energy with respect to nuclear magnetic moments through Coupled-Perturbed Hartree–Fock, CPHF, equations. This is computationally demanding. Therefore, this calculation should be performed only on small, previously optimized geometries.

File: `ethanol_nmr_spinspin.com`

```
%nprocshared=24
%mem=300GB
# b3lyp/6-311+g(d,p) Freq nmr=(giao, spinspin) int=ultrafine

Ethanol NMR with Isotropic Shielding and J-Coupling

0 1
```

The keyword `spinspin` instructs Gaussian to calculate the J-coupling matrix for all atoms.

Extracting the Data

In the `.log` file, scroll beneath the standard shielding tensors until you find the distance-like matrix titled:

```
Total nuclear spin-spin coupling J (Hz):
```

Cross-reference the atom numbers to find the coupling constant between two nuclei. For example, the value crossing a $-CH_3$ proton and a $-CH_2-$ proton gives the vicinal coupling constant:

$$^3J_{\mathrm{HH}}$$

For freely rotating alkyl chains, this value is typically around:

$$^3J_{\mathrm{HH}} \approx 7.0 \text{ Hz}$$

Visualizing the NMR Spectrum: Theory versus Reality

Gaussian outputs idealized "Dirac-delta functions", corresponding to discrete, infinitely thin lines at exact resonance frequencies. Real NMR spectrometers, however, record

Free Induction Decay, FID, signals that relax over time. The Fourier transform of an exponentially decaying signal yields a **Lorentzian line shape**.

To visualize the theoretical spectrum, Gaussian data must be mathematically convoluted with a Lorentzian broadening function:

$$L(x) = \frac{A}{\pi}\left(\frac{\Gamma/2}{(x-x_0)^2 + (\Gamma/2)^2}\right)$$

where x_0 is the chemical shift, A is the peak area, proportional to the number of protons, and Γ is the line width in ppm.

The Ethanol Case Study: CH_3CH_2OH

- $-CH_3$ **group:** This group contains 3 protons. It has 2 neighboring protons from the $-CH_2-$ group and therefore splits into a **triplet**.
- $-CH_2-$ **group:** This group contains 2 protons. It has 3 neighboring protons from the $-CH_3$ group and therefore splits into a **quartet**. It is shifted further to the left because the highly electronegative oxygen atom withdraws electron density, thereby *deshielding* the protons.
- $-OH$ **group:** This group contains 1 proton. In a gas-phase Gaussian calculation at 0 K, this proton may couple with the $-CH_2-$ protons. However, in experimental liquid NMR, rapid intermolecular hydrogen exchange often washes out this coupling, resulting in a broad **singlet**.

What is Next?

I'll upload a program for the elaboration and visualization of NMR spectra. For now, as an exercise, compute the chemical shift for the following molecules:

1. Ethyl acetate
2. Lactic acid
3. Camphor
4. Caffeine
5. o-/m-/p-xylene
6. DMF
7. Toluene or anisole